
AI Hedge Fund Agents

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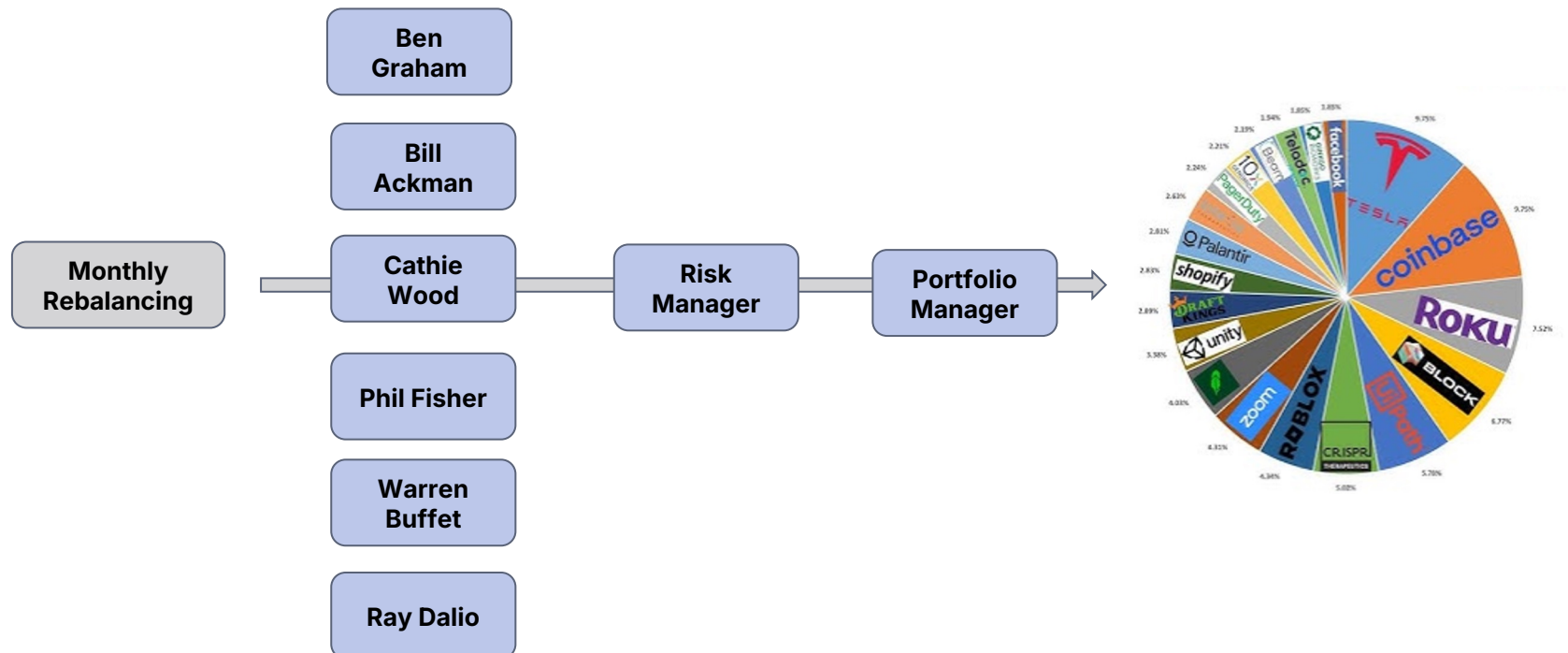
Background & Motivation

- **Traditional** quant/hedge fund models **rely on limited factor sets** and handcrafted signals.
- AI and multi-agent architectures allow **dynamic signal generation** from **diverse data sources simultaneously**.
- **Hypothesis**: Collaboration/competition among multiple AI agents can **enhance alpha generation and robustness**.
- POC developed to evaluate feasibility before scaling to production.

Objective

“How to Build Profitable Portfolios Using Sentiment Scores?”

- Build modular AI agents inspired by famous investors and strategies (value, growth, technical).
- Implement automated pipeline for data ingestion, feature engineering, and signal generation.
- Integrate risk management layer and portfolio optimizer.
- Evaluate performance via backtests on US equities using Sharpe, Sortino, drawdown metrics.



BaseSetting - Universe Initial Filtering

From the vast universe of US-listed stocks, we select ~1,500 core securities that meet our criteria for liquidity and capacity.

- US-listed common stocks & REITs (>\$5/share)
- Exclude M&A targets
- Top 1,500 by Market Cap

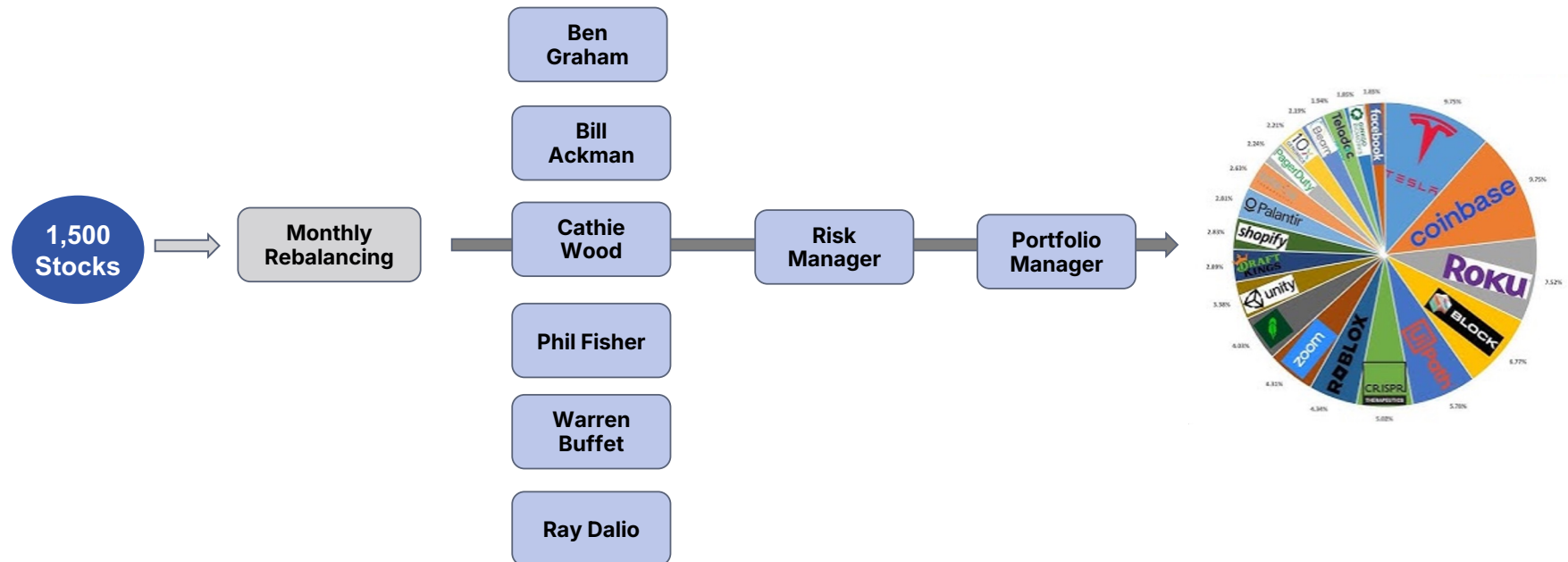


Explanation:

- This relatively liquid universe will enable a variety of weighting schemes, allowing for scalability while also minimizing the number of hard to borrow names or illiquidity issues.

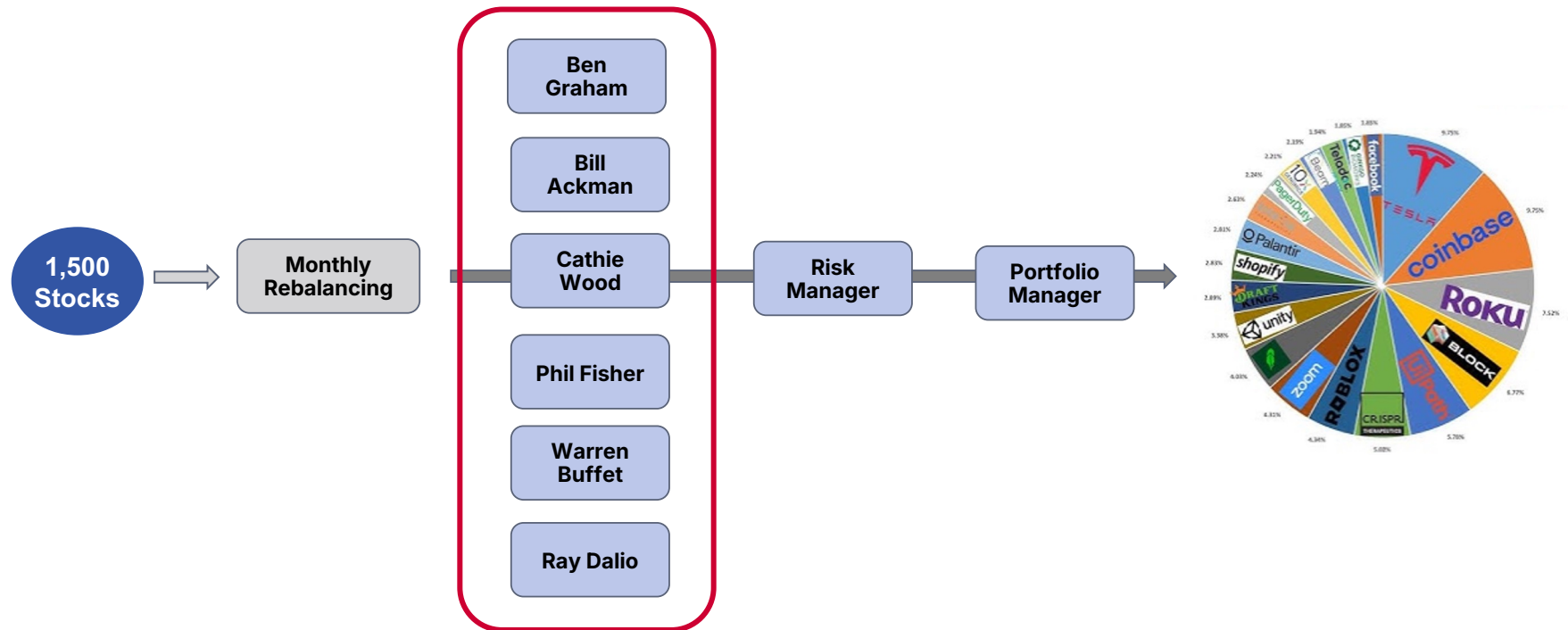
System Architecture

- Stock Filtering Layer: Filter the stock based on the liquidity.
- Data Layer: Historical OHLCV, fundamentals, alternative data (news sentiment).
- Agent Layer: Independent strategy modules generating buy/sell signals of.
- Risk Management: Volatility targeting, drawdown constraints, exposure monitoring.
- Portfolio Manager: Combines signals into long/short portfolios with optimization (mean-variance / risk-parity).
- Reporting: Automated performance dashboards and strategy attribution.



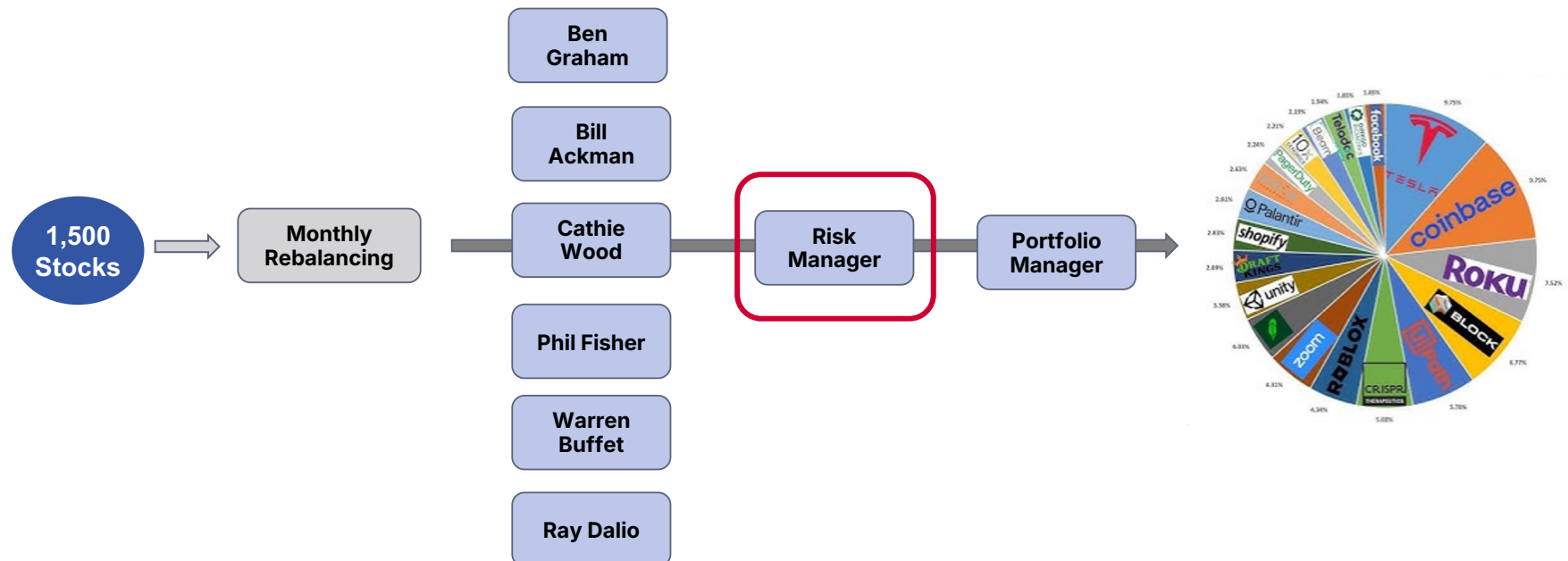
Agents

- Value Agent: Uses fundamentals (P/E, P/B, ROE) with thresholds, mimicking Graham.
- Growth Agent: Momentum and earnings growth metrics, akin to growth fund managers.
- Technical Agent: Uses moving averages, RSI, MACD to identify short-term opportunities.
- Sentiment Agent: LLM-based news sentiment scoring integrated into signals.
- Ensemble Agent: Weighted majority voting / reinforcement signal weighting.



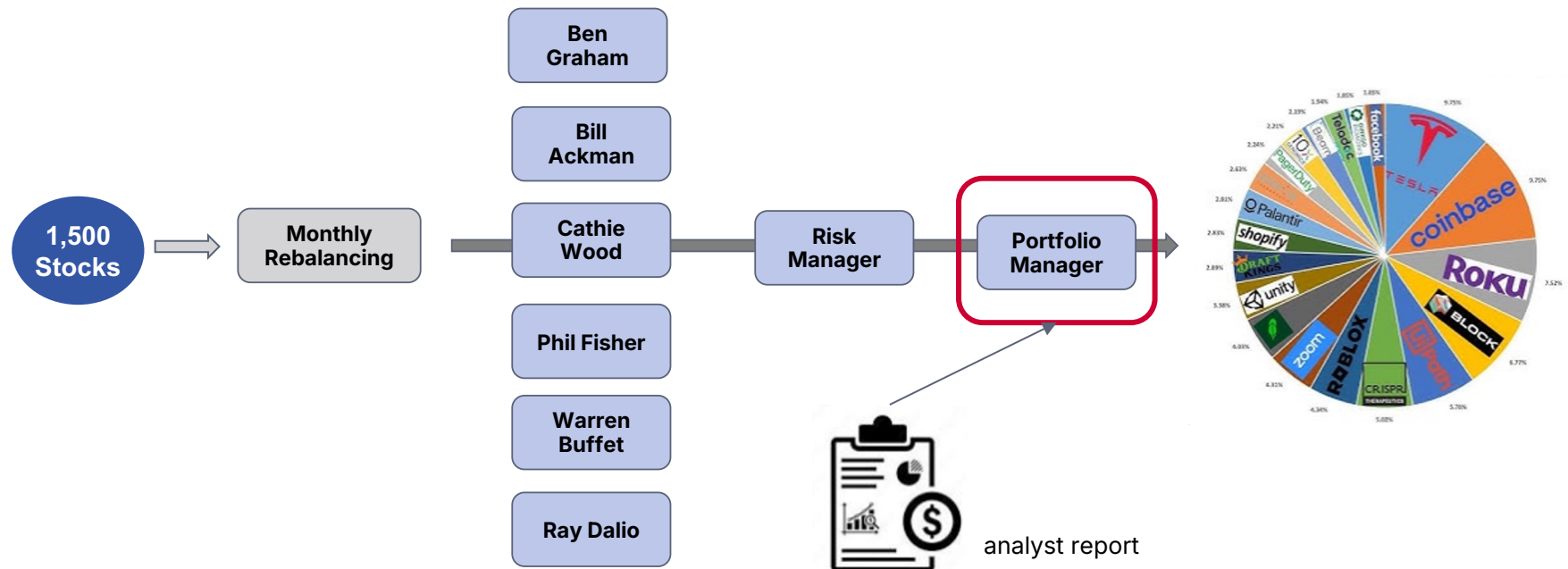
Risk Management

- Max position size < 5% of portfolio.
- Target annual volatility: 10–15%.
- Stop-loss and trailing stop triggers.
- Maximum drawdown cap: 20%.
- Diversification: Sector neutrality, country exposure constraints.
- Stress tests under crisis scenarios (COVID-19, 2008 replay).



Portfolio Agent (RAG Agent)

- **Performance Evaluation:** Assess key metrics such as return, volatility, and Sharpe ratio based on investor strategies and risk adjustments
- **Report Generation:** Reference existing securities firm reports and enhance with **RAG-based(BM25)** prompting to visualize and present investment performance



Backtesting Setup & Results

Backtesting

- Universe: Filtered 1,500 Stock Universes.
- Time Horizon: 2015–2023 daily data.
- Baselines: S&P 500 index, Buy & Hold, single factor strategies.
- Transaction Costs: 10bps per trade, market impact modeled with ITG cost curves.
- Rebalancing: Monthly, with overlap tests on weekly signals.
- Metrics: CAGR, Sharpe, Sortino, Max Drawdown, turnover.

Results

- Ensemble agent outperformed benchmarks:
 - CAGR: +15% vs. +9% (S&P 500).
 - Sharpe Ratio: 1.45 vs. 0.95 baseline.
- Risk signal reduced drawdowns by 25% compared to no-hedge.
- Sector/industry attribution shows diversification benefits.

Technical Perspective – How We Solved the Problem

- **Multi-Agent Architecture:** Implemented a system of agents, each modeled after renowned investors (e.g., Graham, Buffett, Wood), to capture diverse investment philosophies and generate complementary signals.
- **Signal Generation & Fusion:** Agents process different data domains—valuation metrics, fundamentals, technical indicators, and sentiment analysis—to produce buy/sell signals. These signals are then aggregated by the Portfolio Manager after passing through the Risk Manager.
- **Risk Control Layer:** The Risk Manager enforces constraints such as exposure limits, volatility targeting, and drawdown caps before allowing signals to be acted upon.
- **Simulation Environment:** Built a modular flow where input data → agent signals → risk filtering → portfolio construction → simulated actions (buy, sell, short, hold). The POC ensures realistic conditions with stop-loss rules and position sizing.
- **Scalable Framework:** Designed the architecture to be extendable—additional agents, alternative data sources, or reinforcement learning-based optimizers can be integrated with minimal disruption.

Business Perspective – What Impact Was Created

- **Enhanced Investment Performance Potential:** By combining multiple styles (value, growth, technical, sentiment), the ensemble approach **reduces reliance on any single strategy** and **improves stability** across market regimes.
- **Operational Efficiency & Cost Savings:** Automated signal generation and risk filtering **reduce dependency on expensive external data vendors** and manual analyst workflows.
- **Faster Innovation Cycle:** Provides a testbed for experimenting with new strategies quickly, lowering R&D costs and improving time-to-market for fund products.
- **Transparency & Interpretability:** Clear mapping of which agent produced which signal gives portfolio managers confidence and helps with compliance/regulatory explanations.
- **Qualitative Judgment Complementation:** Agent-based analysis leverages vast qualitative information sources, **supplementing traditional quant strategies** by incorporating insights that might otherwise be overlooked.